

Bykea: Getting lives moving in Pakistan's gridlocked megacities

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Bykea gained four customers in four mobility dead miles, halved the number of bikes needed per city, and completed a ratio delivery mission in COVID-19 with Google Maps Platform.

ABOUT BYKEA



About Bykea

Bykea is a Pakistan-based mobility superapp that uses machine learning to empower city dwellers with affordable mobility. It offers ride-hailing, parcel delivery, payment services, and more through its motorbike network, and recently raised \$13 million in Series B financing.

Results:

- Realizes a 20% reduction

Enables average

Industries: Transportation

Location: Pakistan

of "dead miles" where drivers are not providing service

- Enables efficient, affordable mobility by halving the number of drivers needed in a single city
- Enhances social empowerment mission by enabling Urdu language voice commands on app

time
each
job



About MediaAgility

A Google Cloud Premier Partner and US-headquartered Inc. 5000 company, MediaAgility is a global digital consultancy offering full-spectrum digital advisory, transformation, and operations.

Google Maps Platform

(<https://cloud.google.com/maps-platform/>)

Geocoding API

(<https://developers.google.com/maps/documentation/geocoding/intro>)

Google Assistant (<https://assistant.google.com/>)

Distance Matrix

(<https://developers.google.com/maps/documentation/distance-matrix/start>)

Google Places API

(<https://cloud.google.com/maps-platform/places/>)

Opening onto the Arabian Sea, K mix of ethnicities, languages, an home to 18 million people and in the midst of modernizing, which means that getting around this megacity poses a daily challenge (<https://www.bloomberg.com/news/features/2020-11-02/pakistan-s-megacity-tries-to-modernize>)

Seeing the challenges in his beloved city, Muneeb

Maayr and his partners founded Bykea

(<https://www.bykea.com/>), a platform designed to

break the gridlock of urban life through nimble ride-

hailing, on-demand delivery, and versatile courier

services. With 17.5 million bikes

([https://www.riazhaq.com/2019/06/vehicles-in-pakistan-over-half-of-](https://www.riazhaq.com/2019/06/vehicles-in-pakistan-over-half-of-all.html#:~:text=Total%20number%20of%20vehicles%20registered,wagons%20rose%205.3%25%20reaching%203%2C043%2C593.)

[all.html#:~:text=Total%20number%20of%20vehicles%20registered,wagons%20rose%205.3%25%20reaching%203%2C043%2C593.](https://www.riazhaq.com/2019/06/vehicles-in-pakistan-over-half-of-all.html#:~:text=Total%20number%20of%20vehicles%20registered,wagons%20rose%205.3%25%20reaching%203%2C043%2C593.))

on the nation's roads, compared to just three million

cars, the Bykea team saw Pakistan's love of

motorbikes as the way out of the mobility impasse.

By building a network of freelance bikers weaving in

and out of traffic, the team saw an opportunity to

offer fast, affordable, and reliable mobility services

for Karachiites.

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Founding Member & CIO,
Bykea

To make it possible, Bykea needed intelligent mapping tools to enable the startup's proprietary machine learning algorithms to identify the rider best positioned to handle each customer request. It found the solution in Google Maps Platform (/maps-platform), which enabled granular route and distance calculations to match driver flows and mobility requirements with clockwork precision.

"Our mission at Bykea is to empower people in Pakistan's gridlocked cities to get their lives moving," says Abdul Mannan, Bykea Founding Member and CIO. "Through a combination of the Distance Matrix API

(<https://developers.google.com/maps/documentation/distance-matrix/>)

and [Places API](#)

(<https://developers.google.com/maps/documentation/places/web-service/overview>)

, Google Maps Platform provides us tools we need to optimize matches between riders and customers in a dynamic city like Karachi."

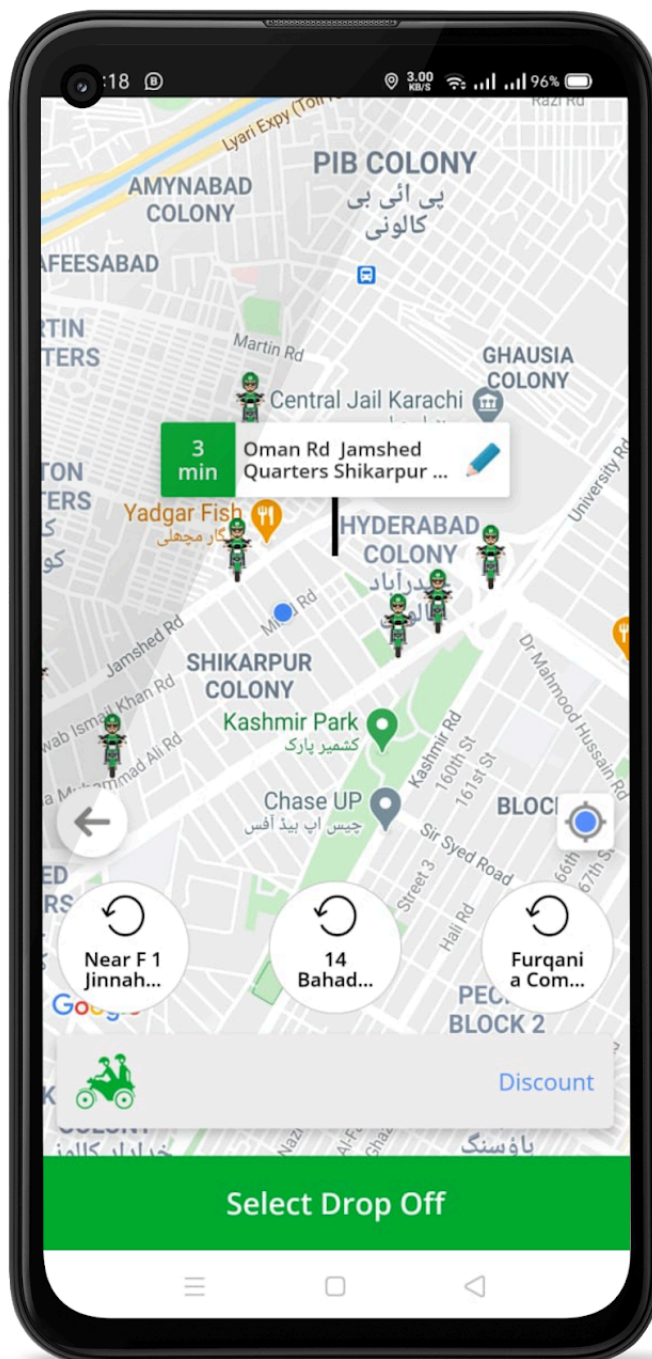
Maximizing the value of each mobility mile with the Distance Matrix API

Since launching in 2016, Bykea has grown into a platform of 30,000 riders serving more than two million customers. Success resides in a network of riders who pick up jobs on the way to wherever they happen to be going anyway—be it work, school, shopping, or leisure. It's the idea "might as well" that makes a transformative difference.

Everyone wins in this system, says Abdul. Drivers make a little extra money (enough for daily groceries) in the natural rhythm of life. Customers benefit from low costs, fast service, and a safe local touch. Bykea benefits by not having to pay people extra to go out of their way to pick up a job.

The environment wins, too. A single itinerary will bundle multiple customer needs, easing traffic congestion and pooling fuel consumption. The rider enjoys the additional benefit of customers shouldering their share of fuel costs for the journey.

It may sound simple. But making the idea work requires an interplay of ML algorithms that require world-class mapping. The Distance Matrix API plays a key role by calculating time and distance to reach a destination, enabling route optimization based on evolving rider-customer parameters.



"If I'm going from point A to point B, and you need to move in the same direction, our ML system will match us together through information provided by the Distance Matrix API," says Abdul. "It's the linchpin of our system that pools mobility needs in huge cities."

In providing mobility to Pakistan's cities, Bykea's main enemy are "dead miles". These are miles that are covered not actively carrying out a job. The Distance Matrix API has enabled Bykea to reduce dead miles by 20% and halve the number of drivers the startup needs to carry out jobs in a single city district. It has also yielded roughly 15% time savings on each transaction.

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—*Nirdesh Chahal* , Head,
Partner Alliances,
MediaAgility

Now Bykea is exploring Google Cloud machine learning solutions such as [AI Platform](#) (/ai-platform) and [AutoML](#) (/automl) to identify ways to further optimize itineraries and reduce dead miles. "Google Cloud machine learning tools definitely can work with Google Maps Platform features to power the evolution of our ML algorithms," says Abdul. "In this way, we foresee endless opportunity for growth and more optimized mobility services on our platform."

Enabling Urdu language mobility with Google Assistant and Place Autocomplete

Bykea's mission is to democratize mobility at all levels of society. One of Pakistan's main obstacles to digital empowerment is language. Although English is an official language of Pakistan, only half of the population are fluent. Instead, Urdu is the

common tongue in a nation of nearly 75 languages. Yet many digital interfaces do not support Urdu text.

Google Maps Platform enables nearly all Pakistanis to access the Bykea app by linking Urdu-enabled voice commands on Google Assistant

(<https://assistant.google.com/>) to the Places API.

Whoever uses Pakistan's lingua franca can speak their address into a mobile device. Place

Autocomplete

(<https://developers.google.com/places/web-service/autocomplete>)

then converts the voice message into text for the

Geocoding API

(<https://developers.google.com/maps/documentation/geocoding/start>)

to transform into coordinates, which the rider uses for optimized pickup or delivery.

Discovering the right partner to drive a social mobility mission

About a year after launching, Bykea found that soaring popularity also brought a need to further optimize its Google Maps Platform service. It turned to the Google Maps Platform team for advice, which led to an introduction to Google Cloud Premier Partner MediaAgility (<https://www.mediaagility.com/>).

MediaAgility worked closely with Bykea to help optimize its API usage costs, by identifying a number of areas where Bykea was using features that were quickly using up credits (and costing money). MediaAgility worked closely with the

Google Maps Platform support team to identify solutions for Bykea.

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When the COVID-19 crisis reached Pakistan, Bykea's partnership with MediaAgility enabled the startup to launch a social mission delivering free food to the poor in major cities. To enable the startup to maximize its humanitarian reach, MediaAgility worked with Google Maps Platform, which was offering additional credits to customers supporting COVID relief, to help Bykea apply for those credits successfully. "Optimizing and helping Bykea in the scaling of their already robust platform on Google Maps was rewarding. Not just because we were working towards increasing efficiencies and decreasing costs, but we also could contribute towards Bykea's social mission during COVID," says Nirdesh Chahal, Head, Partner Alliances, MediaAgility.

"Free credits from Google Maps Platform secured by our partner MediaAgility allowed us to deliver free groceries during COVID-19, at minimum cost," says Abdul. "That meant we didn't have to charge the government delivery fees to bring these groceries to the poor. It's an example of how strategic partnerships are lifting us in our mission to empower Pakistan's biggest cities."